

The Double Pendulum

Lagrangian Theory in the Notation of Taylor, Classical.Mechanics

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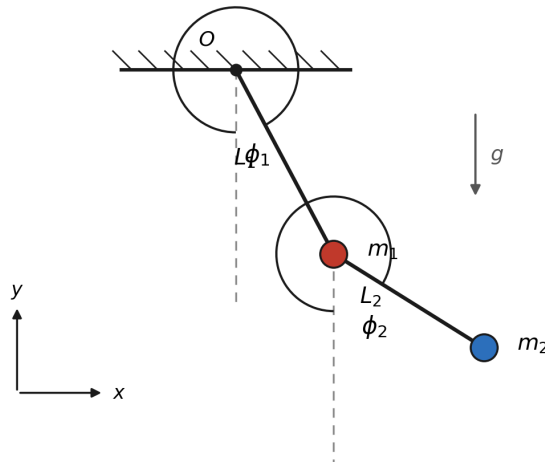
1. The physical system

The **double pendulum** consists of a second simple pendulum hung from the bob of a first. We take two point masses m_1 and m_2 joined by two rigid, massless rods of fixed lengths L_1 and L_2 . The upper rod is suspended from a fixed pivot O , and the whole system swings in a fixed vertical plane under uniform gravity g . Friction at the pivots and air resistance are neglected, so all forces are conservative and the Lagrangian formulation applies directly.

The system has two degrees of freedom. Following Taylor, we choose as generalized coordinates the two angles that each rod makes with the downward vertical,

$$q_1 = \phi_1, \quad q_2 = \phi_2,$$

both measured positive in the same rotational sense. Time derivatives are written with the overdot, $\dot{\phi}_i \equiv d\phi_i/dt$ and $\ddot{\phi}_i \equiv d^2\phi_i/dt^2$.



Schematic of the double pendulum; The generalized coordinates ϕ_1 and ϕ_2 are the angles of each rod from the downward vertical (dashed); The y -axis points upward;

Symbol	Meaning
m_1, m_2	masses of the upper and lower bobs
L_1, L_2	lengths of the upper and lower rods (rigid, massless)
ϕ_1, ϕ_2	angles from the downward vertical (generalized coordinates)
$\dot{\phi}_i, \ddot{\phi}_i$	angular velocity and angular acceleration
g	acceleration due to gravity
$T, U, \mathcal{L}$	kinetic energy, potential energy, Lagrangian
M, K	mass and stiffness matrices (small oscillations)
$\omega_{\pm}$	normal-mode angular frequencies
E	total mechanical energy

2. Generalized coordinates and positions

Place the origin at the pivot O with the y -axis pointing vertically **upward**. The Cartesian positions of the two bobs follow from the geometry:

$$\begin{aligned} x_1 &= L_1 \sin \phi_1, & y_1 &= -L_1 \cos \phi_1, \\ x_2 &= L_1 \sin \phi_1 + L_2 \sin \phi_2, & y_2 &= -L_1 \cos \phi_1 - L_2 \cos \phi_2. \end{aligned}$$

The second bob's position is that of the first plus a displacement of length L_2 along the lower rod, which is why ϕ_1 appears in both coordinates of m_2 . This coupling of the coordinates is the source of all the interesting dynamics.

3. Velocities and the kinetic energy

Differentiating the positions with respect to time gives the velocity components. For the upper bob,

$$\dot{x}_1 = L_1 \dot{\phi}_1 \cos \phi_1, \quad \dot{y}_1 = L_1 \dot{\phi}_1 \sin \phi_1,$$

so that its speed satisfies

$$v_1^2 = \dot{x}_1^2 + \dot{y}_1^2 = L_1^2 \dot{\phi}_1^2.$$

For the lower bob,

$$\dot{x}_2 = L_1 \dot{\phi}_1 \cos \phi_1 + L_2 \dot{\phi}_2 \cos \phi_2, \quad \dot{y}_2 = L_1 \dot{\phi}_1 \sin \phi_1 + L_2 \dot{\phi}_2 \sin \phi_2.$$

Squaring and adding, and using $\cos \phi_1 \cos \phi_2 + \sin \phi_1 \sin \phi_2 = \cos(\phi_1 - \phi_2)$,

$$v_2^2 = L_1^2 \dot{\phi}_1^2 + L_2^2 \dot{\phi}_2^2 + 2L_1 L_2 \dot{\phi}_1 \dot{\phi}_2 \cos(\phi_1 - \phi_2).$$

The total kinetic energy is $T = 1/2 m_1 v_1^2 + 1/2 m_2 v_2^2$, that is,

$$T = 1/2 (m_1 + m_2) L_1^2 \dot{\phi}_1^2 + 1/2 m_2 L_2^2 \dot{\phi}_2^2 + m_2 L_1 L_2 \dot{\phi}_1 \dot{\phi}_2 \cos(\phi_1 - \phi_2)$$

The final cross term, proportional to $\cos(\phi_1 - \phi_2)$, couples the two coordinates and vanishes only when the rods are momentarily perpendicular.

4. The potential energy

With the origin at the pivot and y measured upward, the gravitational potential energy of each bob is $U_i = m_i g y_i$ (taking $U = 0$ at the height of the pivot). Hence

$$U = m_1 g y_1 + m_2 g y_2 = -(m_1 + m_2) g L_1 \cos \phi_1 - m_2 g L_2 \cos \phi_2$$

As expected, U is minimized when both angles are zero, i.e. when the pendulum hangs straight down.

5. The Lagrangian

The Lagrangian is $\mathcal{L} = T - U$:

$$\mathcal{L} = 1/2 (m_1 + m_2) L_1^2 \dot{\phi}_1^2 + 1/2 m_2 L_2^2 \dot{\phi}_2^2 + m_2 L_1 L_2 \dot{\phi}_1 \dot{\phi}_2 \cos(\phi_1 - \phi_2) + (m_1 + m_2) g L_1 \cos \phi_1 + m_2 g L_2 \cos \phi_2.$$

Because $\mathcal{L}$ does not depend explicitly on time, the total energy will be conserved (Section 8).

6. The equations of motion

The Lagrange equations, in Taylor's form, read

$$\frac{\partial \mathcal{L}}{\partial \phi_i} = \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}_i} \right), \quad i = 1, 2.$$

Coordinate ϕ_1 . The generalized momentum and force are

$$\frac{\partial \mathcal{L}}{\partial \dot{\phi}_1} = (m_1 + m_2) L_1^2 \dot{\phi}_1 + m_2 L_1 L_2 \dot{\phi}_2 \cos(\phi_1 - \phi_2),$$

$$\frac{\partial \mathcal{L}}{\partial \phi_1} = -m_2 L_1 L_2 \dot{\phi}_1 \dot{\phi}_2 \sin(\phi_1 - \phi_2) - (m_1 + m_2) g L_1 \sin \phi_1.$$

Carrying out the time derivative and cancelling the common term, then dividing through by L_1 , gives the first equation of motion:

$$(m_1 + m_2)L_1\ddot{\phi}_1 + m_2L_2\ddot{\phi}_2\cos(\phi_1 - \phi_2) + m_2L_2\dot{\phi}_2^2\sin(\phi_1 - \phi_2) + (m_1 + m_2)g\sin\phi_1 = 0$$

Coordinate ϕ_2 . Similarly,

$$\frac{\partial \mathcal{L}}{\partial \dot{\phi}_2} = m_2L_2^2\dot{\phi}_2 + m_2L_1L_2\dot{\phi}_1\cos(\phi_1 - \phi_2),$$

$$\frac{\partial \mathcal{L}}{\partial \phi_2} = m_2L_1L_2\dot{\phi}_1\dot{\phi}_2\sin(\phi_1 - \phi_2) - m_2gL_2\sin\phi_2.$$

After the time derivative, cancellation, and dividing by m_2L_2 , the second equation of motion is

$$L_2\ddot{\phi}_2 + L_1\ddot{\phi}_1\cos(\phi_1 - \phi_2) - L_1\dot{\phi}_1^2\sin(\phi_1 - \phi_2) + g\sin\phi_2 = 0$$

These two coupled, nonlinear, second-order ordinary differential equations completely determine the motion once the initial angles and angular velocities are specified. They are linear in the accelerations $\ddot{\phi}_1, \ddot{\phi}_2$, so they may be solved algebraically for the accelerations and integrated numerically (for example with a Runge–Kutta scheme) to simulate the trajectory.

7. Small oscillations and normal modes

For motion near the stable equilibrium $\phi_1 = \phi_2 = 0$, expand to lowest nontrivial order: $\cos(\phi_1 - \phi_2) \approx 1$, $\sin\phi_i \approx \phi_i$, $\cos\phi_i \approx 1 - 1/2 \phi_i^2$, and drop terms containing products of velocities. The kinetic and potential energies reduce to quadratic forms,

$$T \approx 1/2 (m_1 + m_2)L_1^2\dot{\phi}_1^2 + 1/2 m_2L_2^2\dot{\phi}_2^2 + m_2L_1L_2\dot{\phi}_1\dot{\phi}_2,$$

$$U \approx \text{const} + 1/2 (m_1 + m_2)gL_1\phi_1^2 + 1/2 m_2gL_2\phi_2^2.$$

Writing $\mathbf{q} = (\phi_1, \phi_2)^\top$, these take Taylor's matrix form $T = 1/2 \dot{\mathbf{q}}^\top \mathbf{M} \dot{\mathbf{q}}$ and $U = 1/2 \mathbf{q}^\top \mathbf{K} \mathbf{q}$, with the **mass** (or inertia) matrix $\mathbf{M}$ and the **stiffness** matrix $\mathbf{K}$:

$$\mathbf{M} = \begin{pmatrix} (m_1 + m_2)L_1^2 & m_2L_1L_2 \\ m_2L_1L_2 & m_2L_2^2 \end{pmatrix}, \quad \mathbf{K} = \begin{pmatrix} (m_1 + m_2)gL_1 & 0 \\ 0 & m_2gL_2 \end{pmatrix}.$$

Seeking normal-mode solutions in which both coordinates oscillate at a common frequency, $\mathbf{q}(t) = \mathbf{a} e^{i\omega t}$, substitution into the linearized equations $\mathbf{M}\ddot{\mathbf{q}} + \mathbf{K}\mathbf{q} = 0$ yields the generalized eigenvalue problem

$$(\mathbf{K} - \omega^2\mathbf{M})\mathbf{a} = 0,$$

which has nontrivial solutions only when

$$\det(\mathbf{K} - \omega^2\mathbf{M}) = 0.$$

The two roots $\omega_-^2 < \omega_+^2$ are the squared **normal-mode frequencies**, and the corresponding eigenvectors $\mathbf{a}_\pm$ give the mode shapes.

Equal masses and lengths. For the symmetric case $m_1 = m_2 = m$ and $L_1 = L_2 = L$, the determinant condition reduces to $\omega^4 L^2 - 4\omega^2 gL + 2g^2 = 0$, with solutions

$$\omega_\pm^2 = (2 \pm \sqrt{2}) \frac{g}{L}$$

that is $\omega_-^2 \approx 0.586 g/L$ and $\omega_+^2 \approx 3.414 g/L$. In the **slow mode** (ω_-) the two bobs swing in phase (the same way), while in the **fast mode** (ω_+) they swing in antiphase (opposite ways). Any small-amplitude motion is a superposition of these two modes.

8. Conserved quantities

Since $\mathcal{L}$ has no explicit time dependence, $\partial \mathcal{L} / \partial t = 0$, the energy function (Hamiltonian) is conserved. Moreover the potential U is velocity-independent and T is a homogeneous quadratic form in the velocities, so the conserved energy is simply the total mechanical energy,

$$E = T + U = \text{constant}$$

No coordinate is ignorable (both ϕ_1 and ϕ_2 appear in $\mathcal{L}$), so energy is the only elementary constant of the motion — a first hint that the system is not integrable in general.

9. Chaotic dynamics

Beyond small amplitudes the full equations of Section 6 are genuinely nonlinear and, unlike the single pendulum, the double pendulum is **non-integrable**. For sufficiently large energies its motion is **chaotic**: two trajectories that begin arbitrarily close together diverge exponentially in time, a signature of a positive Lyapunov exponent and sensitive dependence on initial conditions. Practically, this means the long-term trajectory cannot be predicted from finite-precision initial data, even though the equations are perfectly deterministic and energy is exactly conserved. When comparing measured motion with a simulation, agreement can therefore be expected only over a limited time before the inevitable small discrepancies in the initial state grow to dominate.

10. From tracked coordinates to angles

An experiment that records the bob positions frame by frame yields Cartesian coordinates rather than the angles ϕ_1, ϕ_2 that appear in the theory. With the y -axis pointing upward and

the pivot at (X_0, Y_0) , the upper bob at (X_1, Y_1) , and the lower bob at (X_2, Y_2) , the generalized coordinates are recovered from

$$\phi_1 = \text{atan2}(X_1 - X_0, Y_0 - Y_1), \quad \phi_2 = \text{atan2}(X_2 - X_1, Y_1 - Y_2).$$

Here atan2 returns the angle measured from the downward vertical, positive for displacement toward $+x$, consistent with the convention of Section 1. The angular velocities $\dot{\phi}_i$ then follow by numerical differentiation of the recovered angle time series (after suitable smoothing), which allows the measured energy $E = T + U$ and the phase-space trajectory to be reconstructed and compared with theory.